

INFO 5612

Recommender Systems

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	3
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	ML Engineer (100.0 / 100)	74.2 / 100

74	97	+56
CURRENT AVG SCORE across 6 job roles	POTENTIAL SCORE with recommended additions	MAX SCORE GAIN best single-role improvement

CURRENT COURSE DESCRIPTION

Explores the space of personalized information access applications known as recommender systems. This class will introduce students to a range of approaches for building recommender systems including collaborative, content-based, knowledge-based, and hybrid methods. Students will also explore a variety of applications for recommendation including consumer products, music, social media, and online advertising. The course will also examine controversies surrounding recommendation, including Pariser's "filter bubble", and questions of algorithmic bias. Proficiency in Python programming required.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	100.0	100.0	+0.0	LOW
AI Engineer	85.0	100.0	+15.0	MEDIUM
Machine Learning Engineer	79.1	100.0	+20.9	MEDIUM
Data Scientist	72.8	100.0	+27.2	MEDIUM
MLOps Engineer	64.3	82.4	+18.2	MEDIUM
Applied Scientist	43.9	100.0	+56.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Business Communication

2 of 6 roles

insights

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

MEDIUM

Consider Business Communication content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: insights, business.

GAP KEYWORDS BY ROLE

AI Engineer

85.0 → 100.0 (+15.0 pts)

perception

data

models

learning

form

technical

problems

technology

Machine Learning Engineer

79.1 → 100.0 (+20.9 pts)

data

learning

analytics

tools

solutions

exploratory

business

mobile

Data Scientist

72.8 → 100.0 (+27.2 pts)

data

product

analytics

presentation

insights

different

analytical

people

MLOps Engineer

64.3 → 82.4 (+18.2 pts)

data

learning

machine

tools

models

development

Applied Scientist

43.9 → 100.0 (+56.1 pts)

lab

discovery

scientific

computational

research

problems

models

technical

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.